

Zixun Huang

RESEARCH INTERESTS  [zixunh@wharton.upenn.edu](mailto:zixunh@wharton.upenn.edu) —  [Homepage](#) —  [GitHub](#) —  [Google Scholar](#) —  [LinkedIn](#)

Theoretical foundations of machine learning, with an emphasis on **Optimization Theory & Scaling Laws**, **LLM Reasoning & Post-Training & Agentic RL**, and **applications of AI**.

EDUCATION

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| <b>University of Pennsylvania, Wharton School</b><br><i>Ph.D. in Statistics and Data Science</i>                                                           | Aug. 2026 – Present    |
| <b>Peking University, School of Mathematical Sciences</b><br><i>B.S. in Statistics</i> ; Elite Undergraduate Program for Applied Mathematics; GPA: 3.7/4.0 | Sept. 2022 – Jun. 2026 |
| <b>University of California, Berkeley</b><br><i>Visiting Student</i> ; GPA: 4.0/4.0                                                                        | Jan. 2025 – Aug. 2025  |

WORKING EXPERIENCE

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|---------------------------------------------------------------------|-----------------------|
| <b>Tencent HY – Qingyun Talent Program</b> , <i>Research Intern</i> | Jun. 2026 – Aug. 2026 |
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PUBLICATIONS

[1] Jia-Nan Wang\*, **Zixun Huang\***, Kairui Li\*, Lei Wu<sup>†</sup>. *Momentum in Large-Batch Training: Polyak Enlarges the Critical Batch Size, Nesterov Improves Data Efficiency*. **Preprint**.

[2] **Zixun Huang\***, Kishan Panaganti\*, Haitao Mi, Leowei Liang. *FlowSD: Trajectory-Balanced On-Policy Self-Distillation*. **Preprint**. [\[Blog\]](#)

[3] Zhongzhi Li\*, Yucheng Shi\*, Zongxia Li\*, Ruhan Wang, Anhao Li, **Zixun Huang**, Junyao Yang, Lei Ke, Ninghao Liu, Haitao Mi, Leowei Liang. *Recursive Synthesis for Long-Horizon Terminal Tasks*. **Preprint**. [\[Paper\]](#) [\[Hugging Face\]](#) [\[Website\]](#)

[4] **Zixun Huang\***, Jiayi Sheng\*, Zeyu Zheng. *Variance-Aware Baselines and Adaptive Learning Rates for Reinforcement Learning with Verifiable Rewards*. **Under review**. [\[Paper\]](#)

[5] Yuhao Sun, Zekun Wu, **Zixun Huang**, Peijie Zhou. *TracingFlow: A Simulation-Free Trajectory Inference Framework Based on Second-Order Dynamics*. **Under review**.

[6] Binghui Li\*, Fengling Chen\*, **Zixun Huang\***, Lean Wang\*, Lei Wu. *Functional Scaling Laws in Kernel Regression: Loss Dynamics and Learning Rate Schedules*. **NeurIPS 2025 Spotlight**. [\[Paper\]](#) \* Equal contribution.

RESEARCH EXPERIENCE

Optimization Theory & Scaling Laws

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| ► <b>Peking University</b> , <i>Research Intern</i> , Beijing<br><i>Supervisor: Assistant Professor Lei Wu</i>                                                                                                                                                                                                                                                                                                                                                                                                                                               | May. 2023 – Aug. 2026 |
| <ul style="list-style-type: none"><li>• <b>Large-batch momentum [1]</b>: Theoretically characterized how Polyak and Nesterov momentum reshape stochastic-training dynamics; established that Polyak enlarges the critical batch size, while Nesterov improves data efficiency.</li><li>• <b>Functional scaling laws [6]</b>: Developed an intrinsic-time analysis of one-pass SGD that predicts full loss trajectories under arbitrary learning-rate schedules; validated the theory on power-law kernels and 0.1B–1B language-model pre-training.</li></ul> |                       |

LLM Reasoning & Post-Training & Agentic RL

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| ► <b>Tencent Hy LLM – Qingyun Talent Program</b> , <i>Research Intern</i> , Beijing<br><i>Supervisor: Senior Researcher Kishan Panaganti</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Jun. 2026 – Aug. 2026 |
| <ul style="list-style-type: none"><li>• <b>On-policy self-distillation [2]</b>: Developed FlowSD, a trajectory-balance-based OPSD algorithm for learning distributions over complete responses. Across five reasoning benchmarks, improved over GRPO by <b>1.95</b> and <b>2.12 points</b> on Qwen3-4B and Qwen3-8B, while increasing semantic strategy diversity among successful solutions.</li><li>• <b>Long-horizon agent training [3]</b>: Built a recursive synthesis pipeline that evolves verified terminal tasks into harder executable tasks, producing <b>37,484 tasks</b> across 15 rounds at approximately \$0.05 per task; the resulting trajectories improved Qwen3.5 agents through SFT and RL.</li></ul> |                       |
| ► <b>University of California, Berkeley</b> , <i>Research Intern</i> , Berkeley, CA<br><i>Supervisor: Associate Professor Zeyu Zheng</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Jan. 2025 – Aug. 2025 |
| <ul style="list-style-type: none"><li>• <b>Variance-aware policy optimization [4]</b>: Derived a variance-optimal baseline and an SNR-adaptive learning-rate rule for KL-regularized RLVR with convergence guarantees. On Qwen3-4B-Base, OBLR-PO improved average Pass@1 over GRPO by <b>2.52 points</b> across four benchmarks.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                |                       |

SELECTED HONORS AND AWARDS

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|---------------------------------------------|------|
| Grand Prize, Challenge Cup Competition      | 2026 |
| Weiming Scholar, Peking University          | 2026 |
| Hong Sheng Scholarship                      | 2024 |
| First Prize, Chinese Mathematics Contest    | 2024 |
| Yau Contest Groups Prize                    | 2024 |
| Xiaomi Scholarship                          | 2023 |
| Gold Prize, Chinese Mathematical Olympiad   | 2021 |
| Silver Prize, Chinese Mathematical Olympiad | 2020 |

SKILLS

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|---------------------|---------------------------------------------------------------------------------------------|
| <b>Programming:</b> | Python (NumPy, pandas, scikit-learn, PyTorch), C++, MATLAB, L <sup>A</sup> T <sub>E</sub> X |
| <b>Language:</b>    | GRE V160/Q170; GRE Math 970 (95th percentile); TOEFL 101                                    |